

unit: records

1 Search	
50,724 records	
arXiv	11,835
Semantic Scholar	6,744
OpenAlex	14,812
ACL Anthology	1,393
OpenReview	1,754
GitHub	1,361
grey literature	299
citation snowball	12,526
search frozen 2026-09-16	

2 Deduplication	
27,747 unique	
duplicates	22,977
<i>matched on DOI, arXiv id and normalised title</i>	

3 Two-Tier Escalation Screening	
Title and abstract: two votes, tiebreak screened	
	27,747
forwarded to full text	9,967
excluded	17,780
Full text: tier 1, escalation to tier 2	
assessed	8,435
not retrieved	1,532
excluded, with reasons	1,350
7,085 papers included	

the unit changes: included papers are grouped into systems

unit: systems

4 System grouping	
6,504 systems	
from the 7,085 included papers merged on a shared repository, or on equal names where a repository is missing; the census is the sampling frame	

5 Stratified coding frame	
1,234 drawn	
H 984 of 984	w 1
P 150 of 683	w 4.55
O 100 of 4,837	w 48.37
<i>H: ≥ 500 stars, catalogued or vendor, coded in full; P peer-reviewed and O the rest, sampled; w = inverse inclusion probability</i>	

6 Evidence-Anchored Coding	
47,728 cells	
1,256 systems $\times$ 38 dimensions	
■ value + evidence	50.8%
▨ not_reported	48.9%
░ unresolved	0.3%
<i>gate passed: 37 of 38 dimensions at $\kappa \geq 0.6$, 247 systems coded twice</i>	

7 Analyses	
RQ1 taxonomy and its reliability	
RQ2 variation and convergence	
RQ3 design against outcomes, three designs	
RQ4 under-reporting field estimates weighted to the frame; the coded set unweighted	